

MATH 1342 – Elementary Statistics – Week 3: Describing Categorical and Quantitative Variables

Sep 8 – 10 | 2.1 Categorical Variables • 2.2 One Quantitative Variable: Shape and Center | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 6. Analyze data presented in two-way tables to provide information about relationships between categorical variables.	1. Construct frequency and relative frequency tables, compute a sample proportion $\hat{p}$, and distinguish the parameter p from the statistic $\hat{p}$. (CO1) 2. Compute and interpret overall and conditional proportions, and a difference in proportions, from a two-way table, choosing the correct denominator. (CO6) 3. Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CO1, CO6) 4. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CO1) 5. Compute the mean and median of a dataset and explain how skewness and outliers affect each (resistance). (CO2) 6. Interpret the standard deviation as the typical distance from the mean and apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CO2, CO4)	e-Text: 2.1 Categorical Variables	MO1, MO2, MO3	Participation: Frequency and Relative Frequency	MO1	Online Homework: 2.1 Categorical Variables (due Sun Sep 13, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 2.1 Categorical Variables	MO1, MO2, MO3	Participation: Finding Proportions from Two-Way Tables	MO2, MO3		
		Slides: 2.1 Categorical Variables	MO1, MO2, MO3	Participation: Quantitative Variables – shape, mean vs. median	MO4, MO5		
		e-Text: 2.2 One Quantitative Variable	MO4, MO5, MO6	In-class estimation: estimate the mean and SD of a dotplot without calculating (95% rule)	MO6	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 13, 11:59 PM)	MO4, MO5, MO6
		Interactive Lecture: 2.2 One Quantitative Variable	MO4, MO5, MO6				
		Slides: 2.2 One Quantitative Variable	MO4, MO5, MO6				